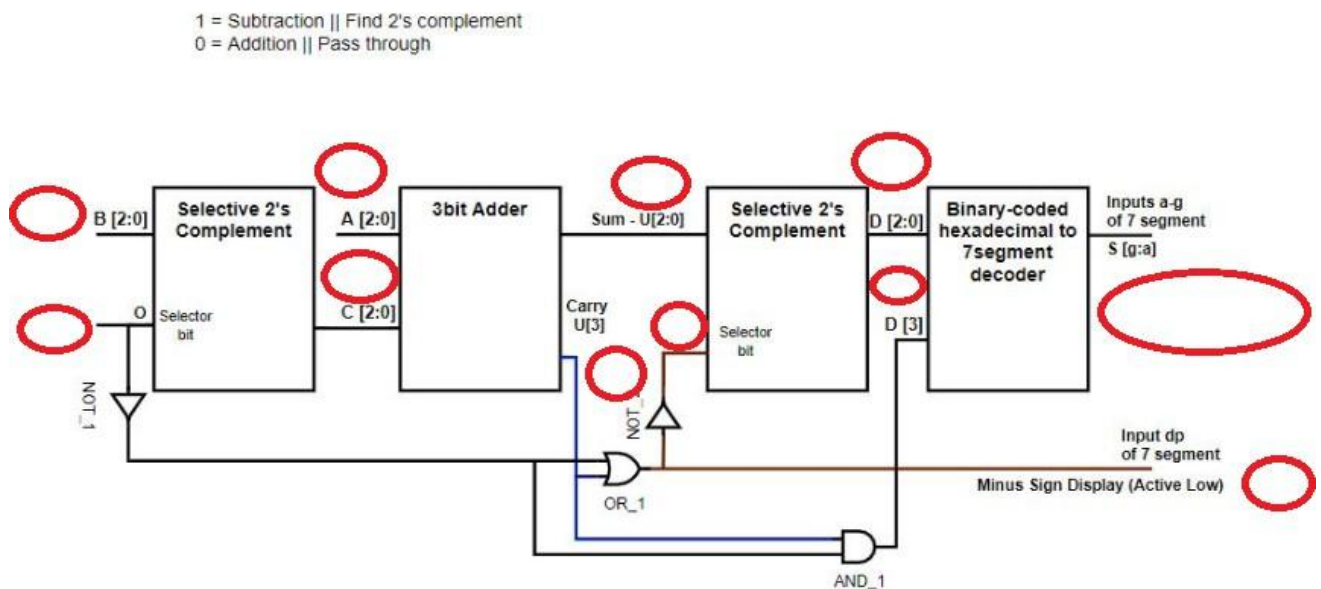


Lab 06 – Worksheet

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a. Introduction

Label the figure below *or* type test cases which you have performed on given design during the lab:



```

module segmentDecoderTest();
    reg [3:0] dD;
    wire [6:0] dS;

    SegmentDecoder(dD, dS); //--UUT

    initial
        begin
            dD = 4'h0;
            #10 dD = 4'h1; // h represents hexadecimal
            #10 dD = 4'h2;
            #10 dD = 4'h3;
            #10 dD = 4'h4;
            #10 dD = 4'h5;
            #10 dD = 4'h6;
            #10 dD = 4'h7;
            #10 dD = 4'h8;
        end

```

```

#10 dD = 4'h9;
#10 dD = 4'hA;
#10 dD = 4'hB;
#10 dD = 4'hC;
#10 dD = 4'hD;
#10 dD = 4'hE;
#10 dD = 4'hF;

$monitor("Time=%0d, D=%4b. S=%7b\n", $time, dD, dS );
end
endmodule

```

b. 7-Segment Hexadecimal Display Decoder

Table 6.1

Hex Digit	Binary-coded Hexadecimal/ Inputs to decoder				Outputs of the decoder						
	D ₃	D ₂	D ₁	D ₀	S _g	S _f	S _e	S _d	S _c	S _b	S _a
0	0	0	0	0	1	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1	0	0	1
2	0	0	1	0	0	1	0	0	1	0	0
3	0	0	1	1	0	1	1	0	0	0	0
4	0	1	0	0	0	0	1	1	0	0	1
5	0	1	0	1	0	0	1	0	0	1	0
6	0	1	1	0	0	0	0	0	0	1	0
7	0	1	1	1	1	1	1	1	0	0	0
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	1	0	0	0	0
A	1	0	1	0	0	0	0	1	0	0	0
B	1	0	1	1	0	0	0	0	0	1	1
C	1	1	0	0	1	0	0	0	1	1	0
D	1	1	0	1	0	1	0	0	0	0	1
E	1	1	1	0	0	0	0	0	1	1	0
F	1	1	1	1	0	0	0	1	1	1	0

Write Boolean expression for each output segment.

Table 6.2

$S_a =$ (1,4,B,D)	$(\sim D_0 \& \sim D_1 \& \sim D_2 \& D_3) \mid (\sim D_0 \& D_1 \& \sim D_2 \& \sim D_3) \mid (D_0 \& \sim D_1 \& D_2 \& D_3) \mid (D_0 \& D_1 \& \sim D_2 \& D_3)$
$S_b =$ (5,6,B,C,E,F)	$(\sim D_0 \& D_1 \& \sim D_2 \& D_3) \mid (\sim D_0 \& D_1 \& D_2 \& \sim D_3) \mid (D_0 \& \sim D_1 \& D_2 \& D_3) \mid (D_0 \& D_1 \& \sim D_2 \& \sim D_3) \mid (D_0 \& D_1 \& D_2 \& \sim D_3) \mid (D_0 \& D_1 \& D_2 \& D_3)$
$S_c =$ (2,C,E,F)	$(\sim D_0 \& \sim D_1 \& D_2 \& \sim D_3) \mid (D_0 \& D_1 \& \sim D_2 \& \sim D_3) \mid (D_0 \& D_1 \& D_2 \& \sim D_3) \mid (D_0 \& D_1 \& D_2 \& D_3)$
$S_d =$ (1,4,7,A,F)	$(\sim D_0 \& \sim D_1 \& \sim D_2 \& D_3) \mid (\sim D_0 \& D_1 \& \sim D_2 \& \sim D_3) \mid (\sim D_0 \& D_1 \& D_2 \& D_3) \mid (D_0 \& \sim D_1 \& D_2 \& \sim D_3) \mid (D_0 \& D_1 \& D_2 \& D_3)$
$S_e =$ (1,3,4,5,7,9)	$(\sim D_0 \& \sim D_1 \& \sim D_2 \& D_3) \mid (\sim D_0 \& \sim D_1 \& D_2 \& D_3) \mid (\sim D_0 \& D_1 \& \sim D_2 \& \sim D_3) \mid (\sim D_0 \& D_1 \& \sim D_2 \& D_3) \mid (\sim D_0 \& D_1 \& D_2 \& D_3) \mid (D_0 \& \sim D_1 \& \sim D_2 \& D_3)$
$S_f =$ (1,2,3,7,D)	$(\sim D_0 \& \sim D_1 \& \sim D_2 \& D_3) \mid (\sim D_0 \& \sim D_1 \& D_2 \& \sim D_3) \mid (\sim D_0 \& \sim D_1 \& D_2 \& D_3) \mid (\sim D_0 \& D_1 \& D_2 \& D_3) \mid (D_0 \& D_1 \& \sim D_2 \& D_3)$
$S_g =$ (0,1,7,C)	$(\sim D_0 \& \sim D_1 \& \sim D_2 \& \sim D_3) \mid (\sim D_0 \& \sim D_1 \& \sim D_2 \& D_3) \mid (\sim D_0 \& D_1 \& D_2 \& D_3) \mid (D_0 \& D_1 \& \sim D_2 \& \sim D_3)$

*Appropriately add comments to your code and **write** the code of your design module in the box below.*

```

module SegmentDecoder(D,S);
    input [3:0] D;
    output [6:0] S;

    // -- S[0]
    assign S[0] = (~D0&~D1&~D2&D3) | (~D0&D1&~D2&~D3) | (D0&~D1&D2&D3) |
    (D0&D1&~D2&D3)

    // -- S[1]
    assign S[1] = (~D0&D1&~D2&D3) | (~D0&D1&D2&~D3) | (D0&~D1&D2&D3) |
    (D0&D1&~D2&~D3) | (D0&D1&D2&~D3) | (D0&D1&D2&D3)

    // -- S[2]

```

```

assign S[2] = (~D0&~D1&D2&~D3) | (D0&D1&~D2&~D3) | (D0&D1&D2&~D3) |
(D0&D1&D2&D3)

// -- S[3]
assign S[3] = (~D0&~D1&~D2&D3) | (~D0&D1&~D2&~D3) | (~D0&D1&D2&D3) |
(D0&~D1&D2&~D3) | (D0&D1&D2&D3)

// -- S[4]
assign S[4] = (~D0&~D1&~D2&D3) | (~D0&~D1&D2&D3) | (~D0&D1&~D2&~D3) |
(~D0&D1&~D2&D3) | (~D0&D1&D2&D3) | (D0&~D1&~D2&D3)

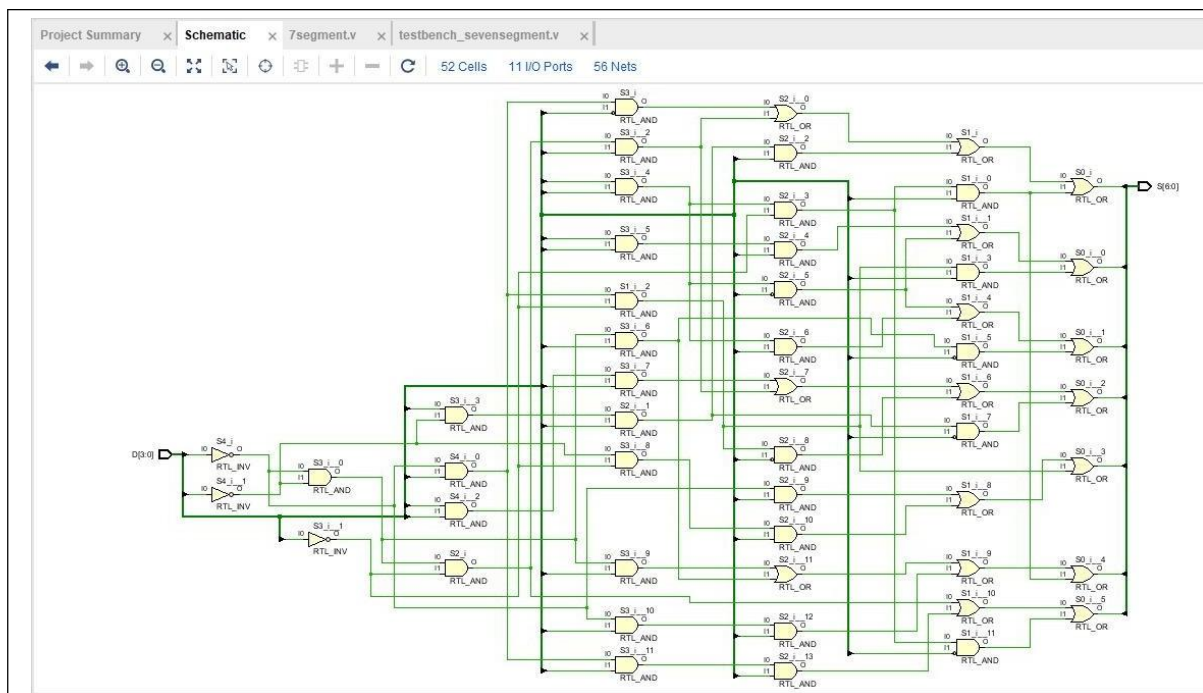
// -- S[5]
assign S[5] = (~D0&~D1&~D2&D3) | (~D0&~D1&D2&~D3) | (~D0&~D1&D2&D3) |
(~D0&D1&D2&D3) | (D0&D1&~D2&D3)

// -- S[6]
assign S[6] = (~D0&~D1&~D2&~D3) | (~D0&~D1&~D2&D3) | (~D0&D1&D2&D3) |
(D0&D1&~D2&~D3)

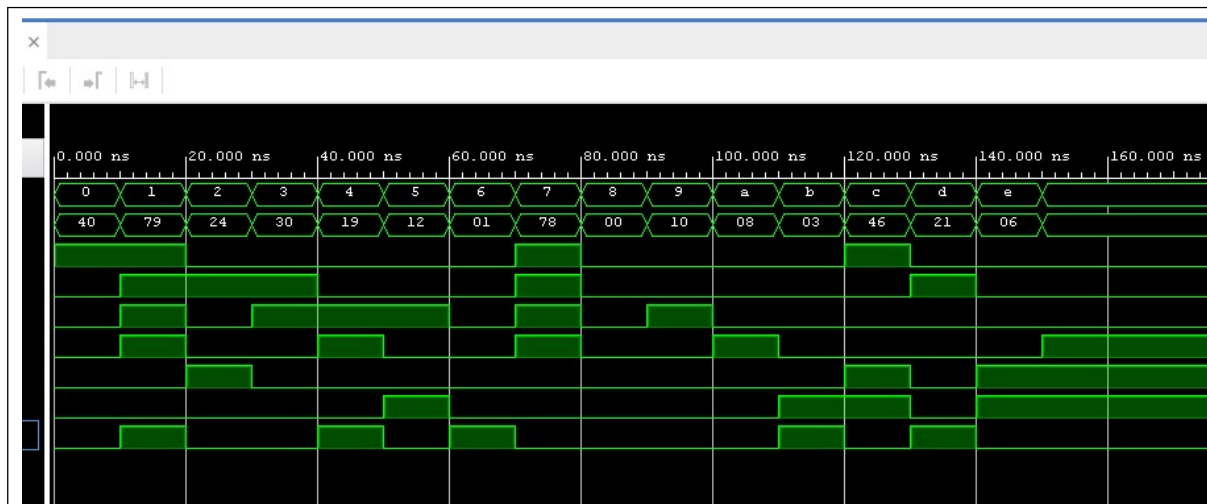
endmodule

```

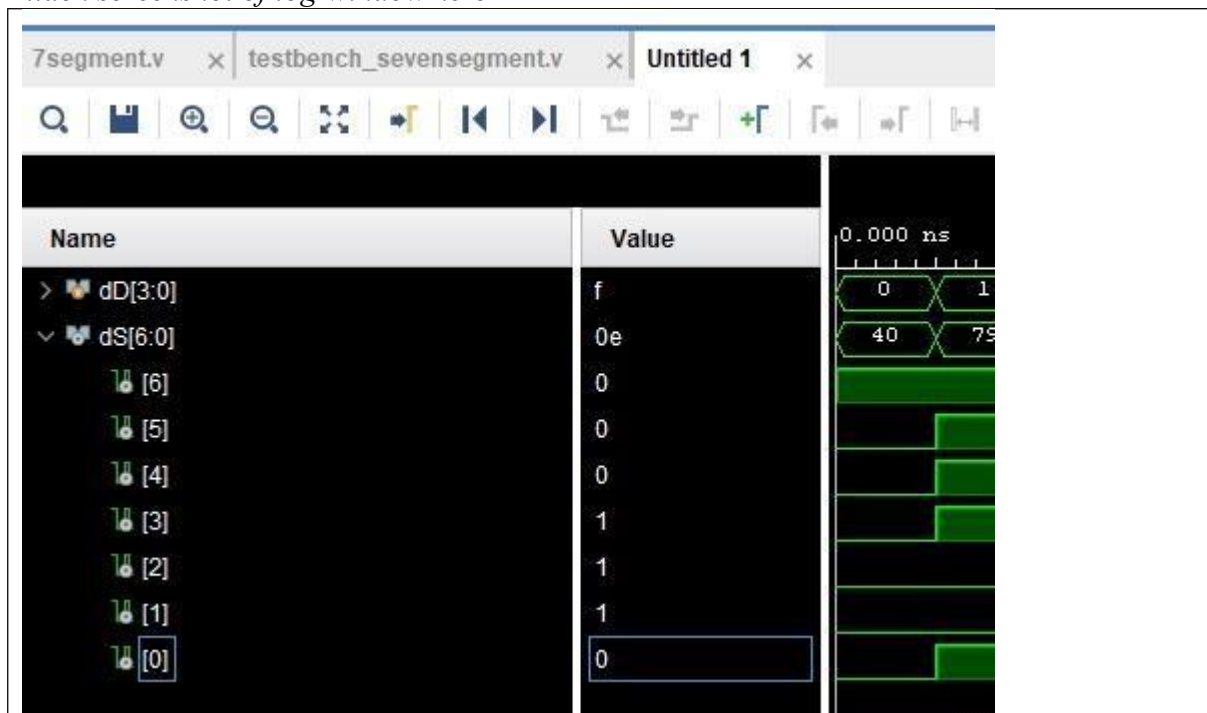
Attach RTL Schematic of the module.



Attach screenshot of your simulation output window: (Invert background color to white using settings option in simulation window)

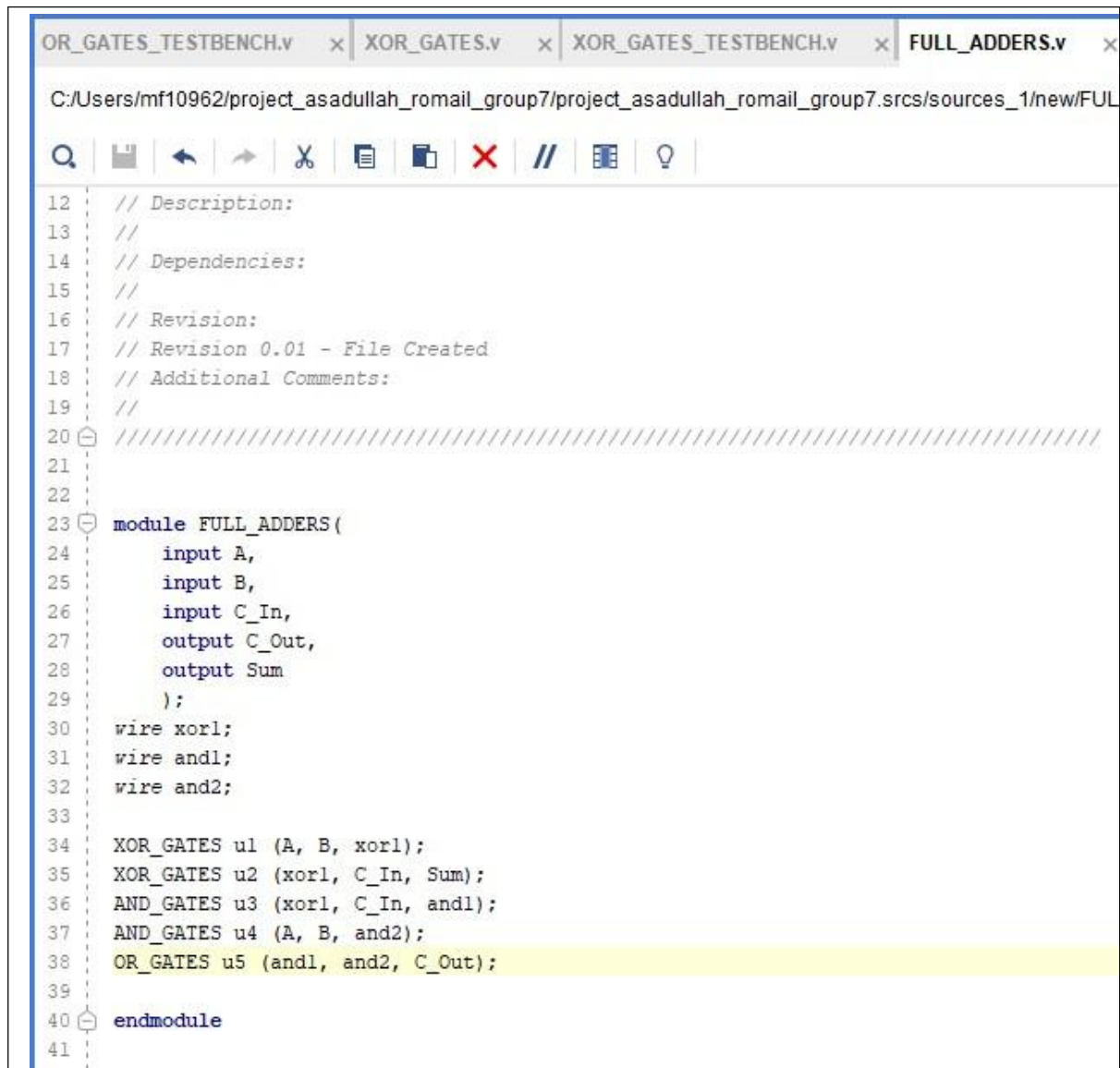


Attach screenshot of log-window here



c. 3-Bit Adder

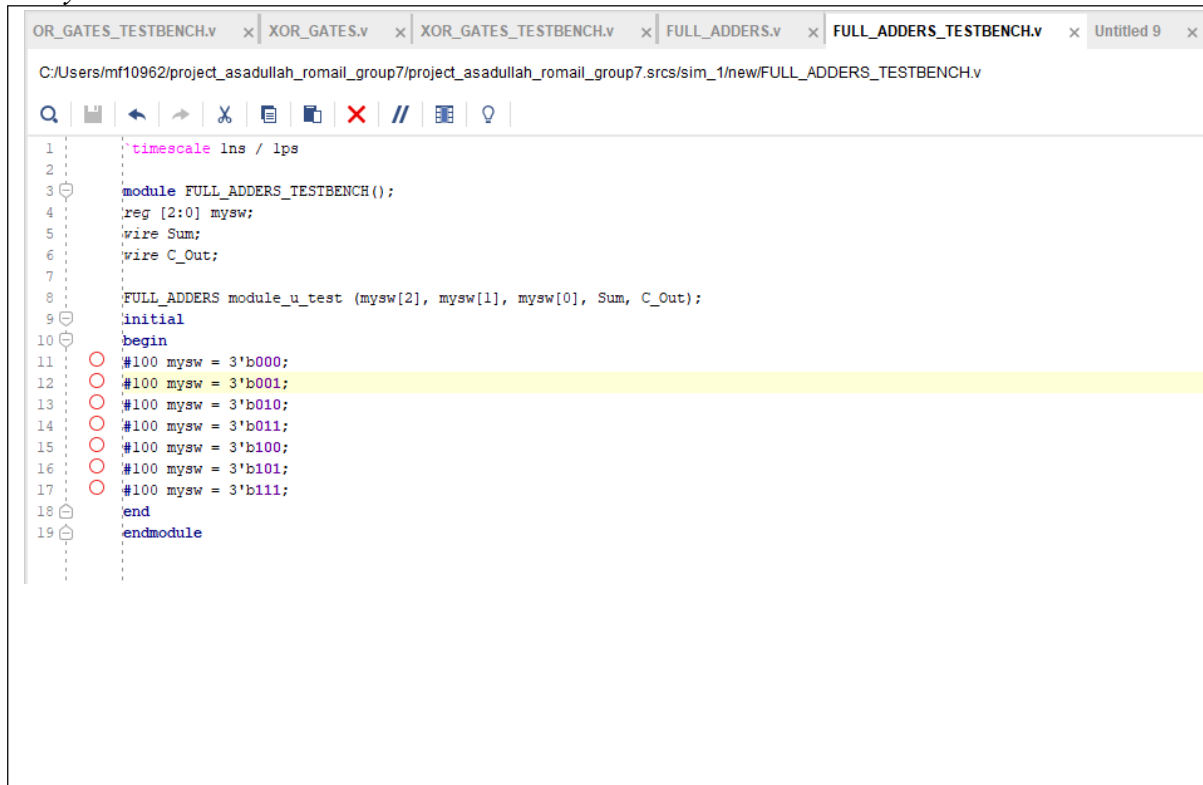
Provide code for design module here



```
OR_GATES_TESTBENCH.v x XOR_GATES.v x XOR_GATES_TESTBENCH.v x FULL_ADDERS.v x
C:/Users/mf10962/project_asadullah_romail_group7/project_asadullah_romail_group7.srscs/sources_1/new/FUL

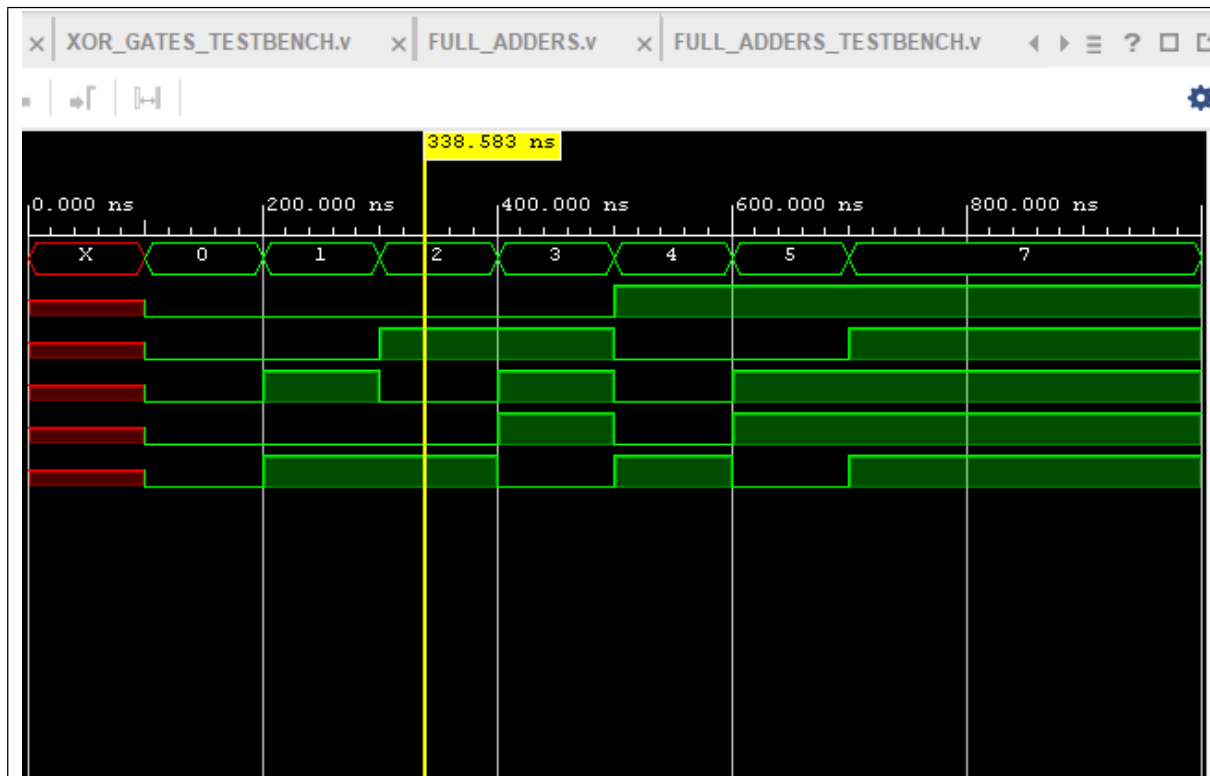
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
21
22
23 module FULL_ADDERS(
24     input A,
25     input B,
26     input C_In,
27     output C_Out,
28     output Sum
29 );
30 wire xor1;
31 wire and1;
32 wire and2;
33
34 XOR_GATES u1 (A, B, xor1);
35 XOR_GATES u2 (xor1, C_In, Sum);
36 AND_GATES u3 (xor1, C_In, and1);
37 AND_GATES u4 (A, B, and2);
38 OR_GATES u5 (and1, and2, C_Out);
39
40 endmodule
41
```

Add your testbench here

The image shows a screenshot of a Verilog testbench for a 3-bit adder. The window title bar shows several files: OR_GATES_TESTBENCH.v, XOR_GATES.v, XOR_GATES_TESTBENCH.v, FULL_ADDERS.v, FULL_ADDERS_TESTBENCH.v, and Untitled 9. The file path is C:/Users/mf10962/project_asadullah_romail_group7/project_asadullah_romail_group7.srcs/sim_1/new/FULL_ADDERS_TESTBENCH.v. The code is as follows:

```
1 timescale 1ns / 1ps
2
3 module FULL_ADDERS_TESTBENCH();
4     reg [2:0] mysw;
5     wire Sum;
6     wire C_Out;
7
8     FULL_ADDERS module_u_test (mysw[2], mysw[1], mysw[0], Sum, C_Out);
9     initial
10    begin
11        #100 mysw = 3'b000;
12        #100 mysw = 3'b001;
13        #100 mysw = 3'b010;
14        #100 mysw = 3'b011;
15        #100 mysw = 3'b100;
16        #100 mysw = 3'b101;
17        #100 mysw = 3'b111;
18    end
19 endmodule
```

Attach screenshot of waveform results here (Invert background color to white using settings option in simulation window)



Attach screenshot of log-window here

The image shows a screenshot of the log window from a simulation. The window has a toolbar with icons for search, save, zoom, and other functions. Below the toolbar is a table with two columns: 'Name' and 'Value'. The table lists the following variables and their current values:

Name	Value
<code>mysw[2:0]</code>	2
<code>[2]</code>	0
<code>[1]</code>	1
<code>[0]</code>	0
<code>Sum</code>	0
<code>C_Out</code>	1

Assessment Rubrics

Marks Distribution:

		LR2 Code	LR4 Data Collection	LR5 Results	LR7 Viva	LR9 Report
In-lab	Task a				10 points	10 points
	Task b	15 points	20 points	15 points		
	Task c	20 points		10 points		
Total Marks =		100 points				

Marks Obtained:

		LR2 Code	LR4 Data Collection	LR5 Results	LR7 Viva	LR9 Report
In-lab	Task a	-	-	-		
	Task b					-
	Task c		-			-
Total Marks =						